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**ADANI UNIVERSITY | DEMMISTO TECHNOLOGIES PVT. LTD.**

## CYBER SAMVAD

# HACKATHON 2026

### Problem Statements & Student Guidelines

**Build a practical solution. Explain the problem clearly. Show how your idea can help real people.**

## Start Here

This guide explains the six hackathon problem statements in simple language. Read it with your whole team before you start. You do not need to build a complete commercial product. Build a working prototype that clearly demonstrates your idea, how it works, and how it could be improved.

### Team Rules

- Each team must have a minimum of 2 and a maximum of 4 students.
- Choose one team name and one team leader. The leader coordinates communication and submits the team details.
- Every team member should contribute and be able to explain the project.
- Select or confirm your assigned problem statement as instructed by the organisers.
- Use sample, synthetic, or publicly permitted data. Do not collect real people's sensitive data without permission.

### What every team should prepare

- A clear explanation of the problem and who is affected.
- A simple working prototype, mock-up, app, dashboard, or simulation.
- A short explanation of the solution and its main features.
- A demo using sample data that shows the solution in action.
- A short presentation covering the problem, solution, impact, limitations, and next steps.

# The Six Problem Statements

Choose the problem your team has been assigned. The suggested features below are starting points—not a compulsory checklist.

# 1. “Is This Video Real?” — Indian Deepfake Detection Platform

*Help people check whether a video may have been manipulated or generated using AI.*

**The challenge:** Deepfake videos can spread misinformation, impersonate people, and cause harm. Create a tool that gives users useful warning signs before they trust or share a video.

## What your solution could include

- Let a user upload a sample video or select a demo file.
- Show a result such as “Possible manipulation detected”, “No strong signs detected”, or “Unable to assess”.
- Highlight clues where possible, such as unusual facial movement, audio-video mismatch, or suspicious frames.
- Show a confidence or risk level with a plain-language explanation.
- Include a reminder that automated results can be wrong and should not be treated as final proof.

## What to demonstrate

- Run the prototype on sample videos, including at least one normal video and one manipulated/demo example if available.
- Show the result and explain why the tool raised a warning.
- Explain what the system cannot reliably detect.

**Important: Do not label a real person as a liar or claim a video is definitely fake based only on an AI score. Use consented or openly licensed sample media.**

# 2. UPI Fraud Prevention — “Stop Fraud Before Payment”

*Warn users about suspicious payment requests before they send money through UPI.*

**The challenge:** People may be tricked by fake collect requests, QR-code scams, impersonation, or pressure to share a PIN. Design a tool that helps users pause, verify, and make safer choices.

## What your solution could include

- Create a simple risk-check screen for a sample payment request, QR code, message, or link.
- Flag warning signs such as urgent language, unknown recipient, suspicious links, or requests to enter a UPI PIN to receive money.
- Give a clear action: “Pause and verify”, “Do not share your PIN”, or “Check recipient details”.
- Provide a short fraud-awareness checklist in simple language.
- Use sample data; do not connect to real bank accounts or payment systems.

## What to demonstrate

- Test the tool with one normal example and at least two suspicious examples.
- Show which warning signs triggered the alert.
- Explain how the tool helps users avoid a risky action.

**Important: Never ask for or store real UPI PINs, OTPs, passwords, bank details, or payment credentials. The prototype must not initiate real payments.**

# 3. Smart Government Hospital Queue Management

*Reduce confusion and waiting-time uncertainty for patients at government hospitals.*

**The challenge:** Crowded registration counters and unclear queues can make hospital visits stressful. Build a simple system that helps patients understand their token, queue status, and likely waiting time.

## What your solution could include

- Generate a demo token number for a patient or department.
- Display current token, approximate waiting time, and queue status on a dashboard.
- Allow staff to update the queue using a simple interface.
- Show separate queues for sample departments such as OPD, pharmacy, or laboratory.
- Include an accessible design with clear text and support for patients who may not be comfortable with technology.

## What to demonstrate

- Create a few sample patients and show how the queue changes when a token is served.
- Demonstrate the patient view and staff view.
- Explain how the design could reduce confusion without delaying urgent cases.

**Important: This is a queue-support prototype, not a medical decision system. Emergency cases must follow hospital triage and staff instructions.**

# 4. Smart City and Critical Infrastructure Cybersecurity

*Help city operators identify and respond to cyber risks affecting important services.*

**The challenge:** City services such as water, traffic, public lighting, and transport depend on connected systems. Create a safe dashboard or simulation that helps an operator notice suspicious activity and prioritise response.

## What your solution could include

- Build a mock dashboard for sample assets such as traffic signals, water sensors, or public-service servers.
- Display simulated alerts, asset status, and a simple risk level.
- Prioritise alerts by potential impact and urgency.
- Provide a basic response checklist, such as isolate a test device, notify the responsible team, and record the incident.
- Include an incident log showing alert time, status, and action taken.

## What to demonstrate

- Trigger a simulated suspicious event and show how it appears on the dashboard.
- Demonstrate how the operator prioritises and records the incident.
- Explain how the prototype supports resilience and continuity of public services.

**Important: Use only a simulated environment or systems you are authorised to test. Do not scan, access, disrupt, or attack real city infrastructure.**

# 5. Smart Darshan & Crowd Management System

*Help visitors plan darshan and help organisers manage crowd flow safely.*

**The challenge:** During busy periods, visitors may face long queues, congestion, and difficulty finding entry or exit routes. Build a tool that gives useful crowd information and helps staff respond to congestion.

## What your solution could include

- Show a simple map or layout with entry, exit, queue, and waiting areas.
- Display simulated crowd levels such as Low, Medium, or High.
- Provide estimated waiting time based on sample queue data.
- Suggest a safer route or waiting area when a zone becomes crowded.
- Provide an organiser dashboard with alerts for high-density zones.

## What to demonstrate

- Use a sample layout and change the crowd level in one zone.
- Show how the visitor information and organiser alert update.
- Explain how the system avoids sending more people into a crowded area.

**Important: Do not use real-time face recognition or identify individual devotees. Use simulated counts or privacy-preserving, non-identifying data.**

# 6. AI-Powered Smart Temple Management & Devotee Safety System

*Support temple operations and devotee safety through useful digital services.*

**The challenge:** Temples may need to coordinate facilities, announcements, lost-and-found requests, accessibility support, and safety alerts. Build a practical system that helps staff and visitors get the right information quickly.

## What your solution could include

- Create a visitor information page with timings, facilities, accessibility details, and announcements.
- Provide a simple lost-and-found or help-request workflow using demo entries.
- Create a staff dashboard for safety notices, maintenance issues, or crowd-related alerts.
- Use AI only where it adds clear value, such as a question-answer assistant based on approved temple information.
- Show status updates so staff can track whether a request has been handled.

## What to demonstrate

- Submit a sample help request and show how staff receive and update it.
- Ask the information assistant a question covered by the demo information.
- Explain how the system improves service while respecting privacy and religious sensitivity.

**Important: Use respectful language and approved information. Do not collect unnecessary personal or religious data, and do not invent official temple guidance.**

# How to Build a Strong Hackathon Prototype

Keep the idea focused. A small feature that works clearly is better than many features that do not work.

1. Step 1 — Understand: Write the problem in your own words and identify who needs the solution.
2. Step 2 — Plan: Decide the 2–4 most important features. Sketch the screen or workflow before coding.
3. Step 3 — Build: Create a working demo using tools and technologies your team can manage.
4. Step 4 — Test: Try normal and edge cases. Fix confusing screens and obvious errors.
5. Step 5 — Prepare: Make sure the demo, sample data, and presentation are ready and can run reliably.
6. Step 6 — Explain: Tell the jury what problem you solve, how it works, who benefits, and what you would do next.

## Suggested Final Presentation

- Problem and users: What is the issue and who experiences it?
- Solution: What have you built, and why is it useful?
- Live demo: Show the main workflow using sample data.
- Impact: How could it improve safety, access, trust, efficiency, or awareness?
- Responsible design: What are the privacy, security, fairness, or safety limits?
- Next steps: What would you improve if you had more time?

## Simple Evaluation Guide

| What the jury may look for | What a strong team shows |
|---|---|
| Problem understanding | The team explains the real problem and who is affected. |
| Practical solution | The idea is useful, focused, and suitable for the stated problem. |
| Working prototype | The main workflow can be demonstrated clearly. |
| Innovation and impact | The solution adds value and could help people or organisations. |
| Responsible design | The team considers privacy, security, safety, and limitations. |
| Team explanation | All members understand their contribution and can answer questions. |

## Final Checklist

- ☐ Our team has 2–4 students and one team leader.
- ☐ We understand the assigned problem statement.
- ☐ Our prototype works for at least one clear use case.
- ☐ We are using safe sample data and authorised tools only.
- ☐ We can explain the main features, limitations, and real-world benefit.
- ☐ Every team member is ready to explain their contribution.

**Remember: The goal is not just to present an idea. Show a clear problem, a practical solution, and a working demonstration.**

## All the best, teams!

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